Assignment # .....

Instructions: .....

1. (DISCx25x08x15x009)
   topics: DISC PROPLOGIC
   How many rows appear in a truth table for each of these compound propositions?

   (a) $X \vee \neg X$

   (b) $A \to (A \wedge \neg B)$

   (c) $(P \vee \neg Q) \leftrightarrow ((P \wedge Q) \to \neg P)$

   (d) $(A \vee B) \to (A \vee B \wedge \neg C)$

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2. (DISCx25x08x15x010)
   topics: DISC PROPLOGIC
   Construct a truth table for the following proposition: $X \vee \neg X$

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3. (DISCx25x09x11x001)
   type: TEST , topics: DISC PROPLOGIC
   Construct a truth table for the following proposition and state whether it is a tautology, contradiction, or contingency: $(X \wedge \neg Y) \to (\neg Y)$

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4. (QUANTx25x08x18x007)
   topics: DISC QUANT
   Determine the truth value of each of these statements if the domain consists of all rational numbers. In both cases, what if we changed the domain to be just the integers?

   (a) $\forall x \exists y \; x \cdot y = 1$

   (b) $\forall x (x > 0 \to (x - 3) > 0)$

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5. (QUANTx26x02x04x001)
   type: HW , topics: DISC QUANT
   Determine the truth value of each of these statements if the domain consists of all integers

   (a) $\exists u \forall x \; x < \sqrt{u}$

   (b) $\forall x \exists u \; x < \sqrt{u}$
